ST10106989

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PDNA Part 2 report

The purpose of this study was to develop and evaluate a python based classification model that predicts whether a breast tumor is benign or malignant. This relates the medical scheme scenario because a reliable classification process could help identify cases that require further clinical attention. The study uses exploratory data analysis (EDA), data preparation, model training, cross validation and test set evaluation to compare several classification algorithms. The target variable is diagnosis with B representing benign and M representing malignant. This is a machine learning exercise using a supplied enhanced data set and it is not a clinical diagnostic tool and should not be used to make decisions about rural patients without appropriate clinical validation.

DATASET SUITABILITY AND QUALITY

The CSV contains 5500 records and 17 columns. This is suitable for supervised binary classification because each record has a known class label and numerical measurements from which a model can learn patterns. The target contains 3470 Benin records (63.1%) and 2030 malignant records (36.9%). The data set therefore has both classes with moderate imbalance. The initial checks that I did found no missing values and no exact duplicate rows. The ID column was excluded from predictors because it is an identifier and not a tumor measurement. A limitation is that the source, collection process and formulas for engineered columns such as malignancy_*risk*_score, tumor_aggressiveness and interaction features are not documented in the supplied materials. These fields must be checked for target leakage, that is, if they were derived from diagnosis or information unavailable at prediction time as model scores may be over optimistic.

ANALYSIS PLAN AND METHODOLOGY

Stage	Approach
EDA and cleaning	Inspect dimensions, data types, descriptive statistics, missing values, duplicate, distributions, outliers, class balance, and feature correlations.
Preparation	Encode diagnosis, remove ID from predictors and use a stratified 80/20 train test split.
Model training	Compare logistic regression, random forest, extra trees, gradient boosting and XGBoost.

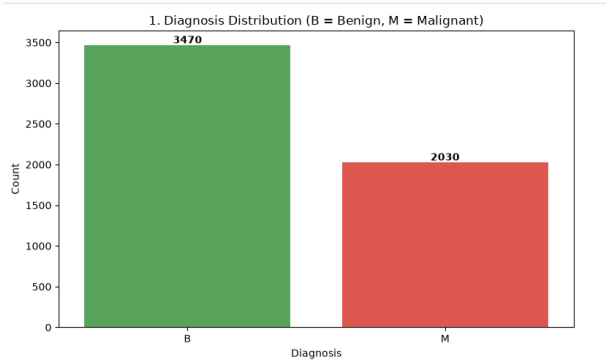
Evaluation	Use five-fold cross validation, accuracy, precision, recall, F1 score, confusion matrix and ROC-AUC.
Improvement and reporting	Compare models, inspect feature importance, discuss errors and limitations and identify further tuning and leakage checks.

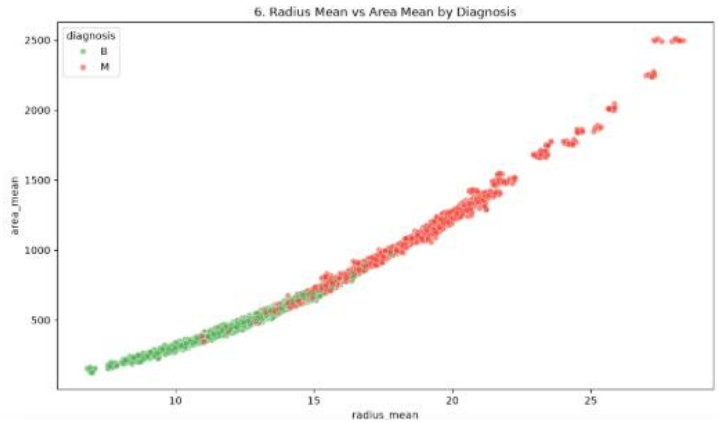
The split used 4400 records for training and 1100 for testing with stratification to preserve the class proportions. A random state of 42 supported reproducibility. Cross validation was applied to the training data, leaving the test set for final evaluation.

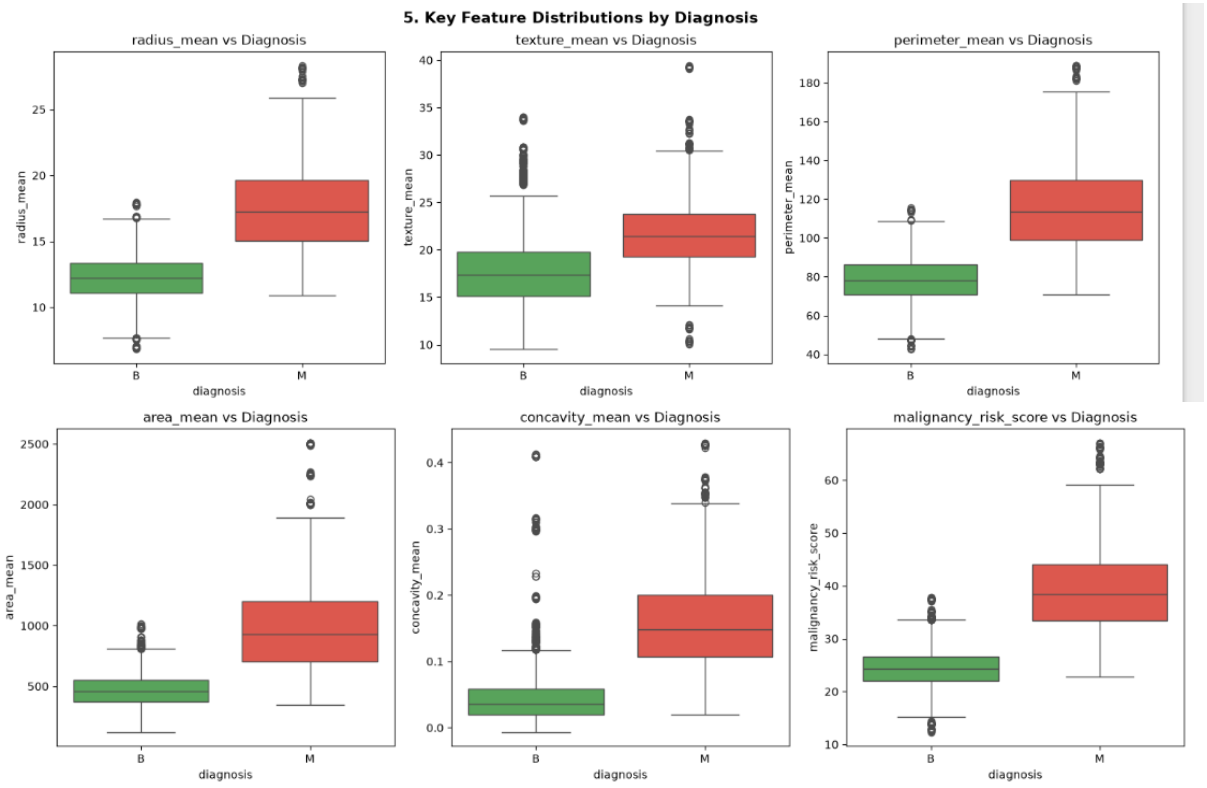
EXPLARATORY DATA ANALYSIS

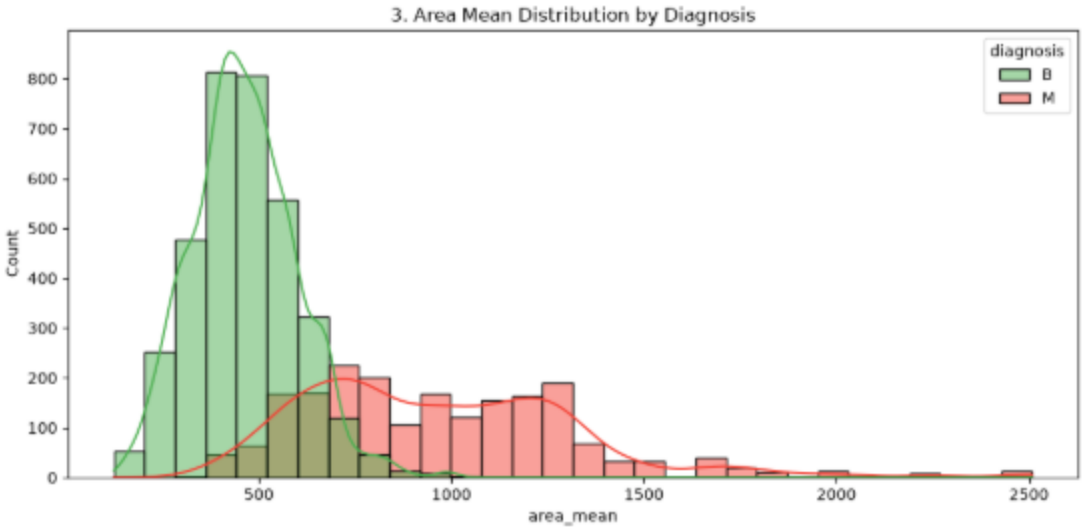
The data inspection confirm 5500 rules in 17 columns with no missing values or exact duplicates. The diagnosis chart showed that benign cases were more common than malignant cases, so accuracy alone is not sufficient to evaluate models. The radius, perimeter, and area plots showed that malignant cases generally had larger size measurements. Previous in area were strongly related, meaning some features may contain overlapping information. Concavity and compactness tended to be higher for malignant cases, while shape_irregulatory and border_complexity also showed differences between classes. However, the overlap between classes means individual features cannot perfectly distinguish every case. The correlation, heatmap and pairplot helped identify relationships among predictors. Engineered risk-related features appeared strongly associated with diagnosis, but their formulas are unknown. They should be reviewed for possible leakage before the results are trusted.

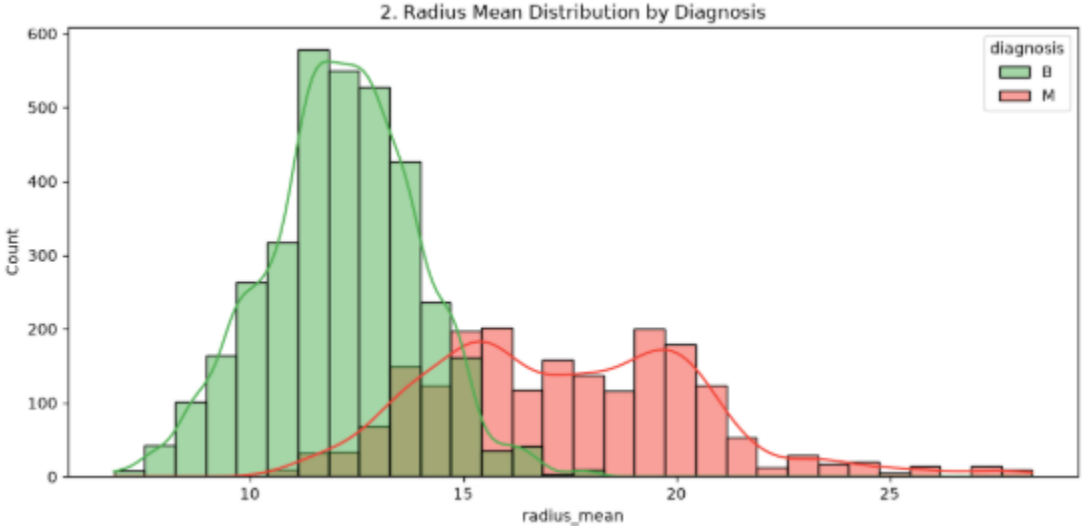
EDA FIGURES

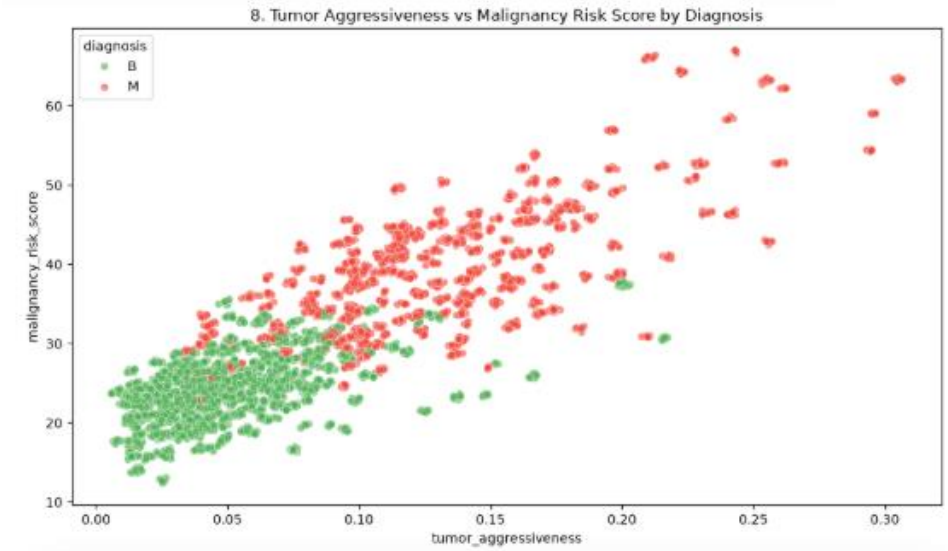



6. Radius Mean vs Area Mean by Diagnosis
diagnosis
B
M
area_mean
radius_mean
2500
2000
1500
1000
500
10
15
20
25


5. Key Feature Distributions by Diagnosis
radius_mean vs Diagnosis
texture_mean vs Diagnosis
perimeter_mean vs Diagnosis
area_mean vs Diagnosis
concavity_mean vs Diagnosis
malignancy_risk_score vs Diagnosis
diagnosis
B
M


3. Area Mean Distribution by Diagnosis
diagnosis
B
M
Count
area_mean
0
100
200
300
400
500
600
700
800
500
1000
1500
2000
2500


2. Radius Mean Distribution by Diagnosis
diagnosis
B
M
Count
radius_mean
0
100
200
300
400
500
600
10
15
20
25


8. Tumor Aggressiveness vs Malignancy Risk Score by Diagnosis
diagnosis
B
M
malignancy_risk_score
tumor_aggressiveness
10
20
30
40
50
60
0.00
0.05
0.10
0.15
0.20
0.25
0.30

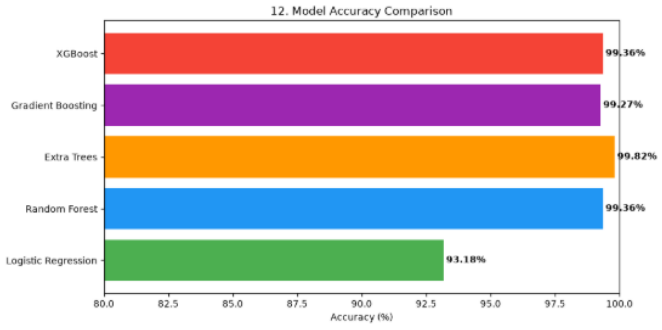
MODEL DEVELOPEMENT AND EVALUATION

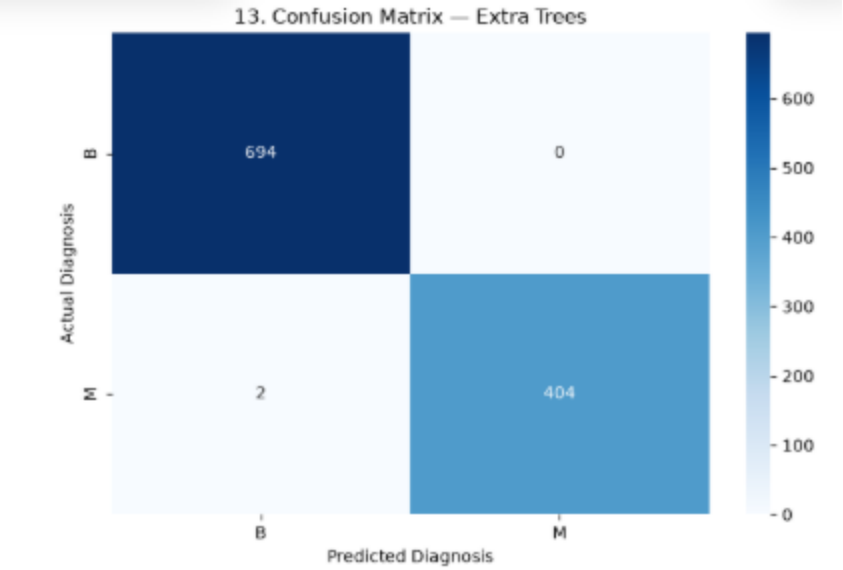
I got to compare 5 algorithms. Logistic regression provided a baseline. Command. Random forest and extra trees can model nonlinear patterns through ensembles of trees. Gradient boosting an XGBoost build trees sequentially to improve predictions. Logistics recreation uses standard scalar and max_iter = 1000. Random forests and extra trees use 200 trees. Current Boosting and XGBoost used 250 estimators, a learning rate of 0.05 and maximum depth of 4.

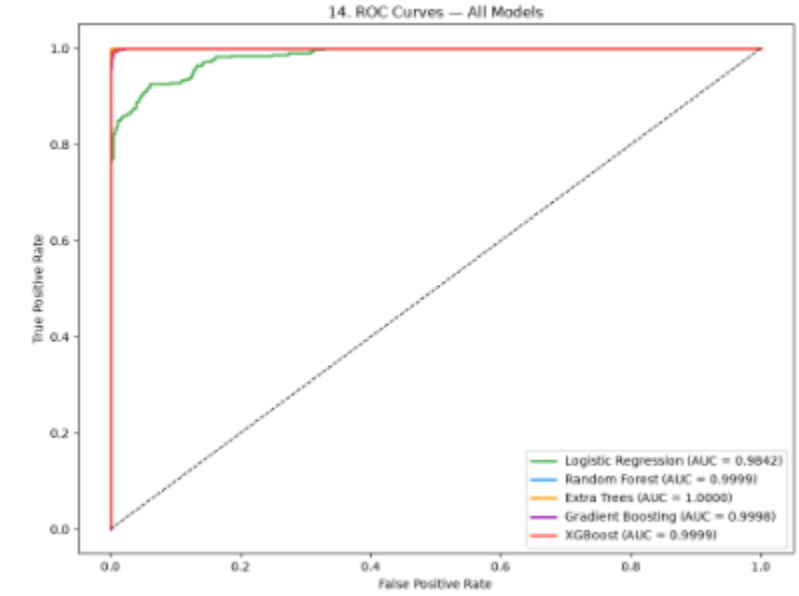
Model	Test accuracy	ROC-AUC	Mean 5-fold CV accuracy
Logistic regression	93.18%	0.9842	94.20%
Random forest	99.36%	0.9999	99.61%
Extra trees	99.82%	1.0000	100.00%
Gradient boosting	99.27%	0.9999	99.52%
XGBoost	99.36%	0.9999	99.59%

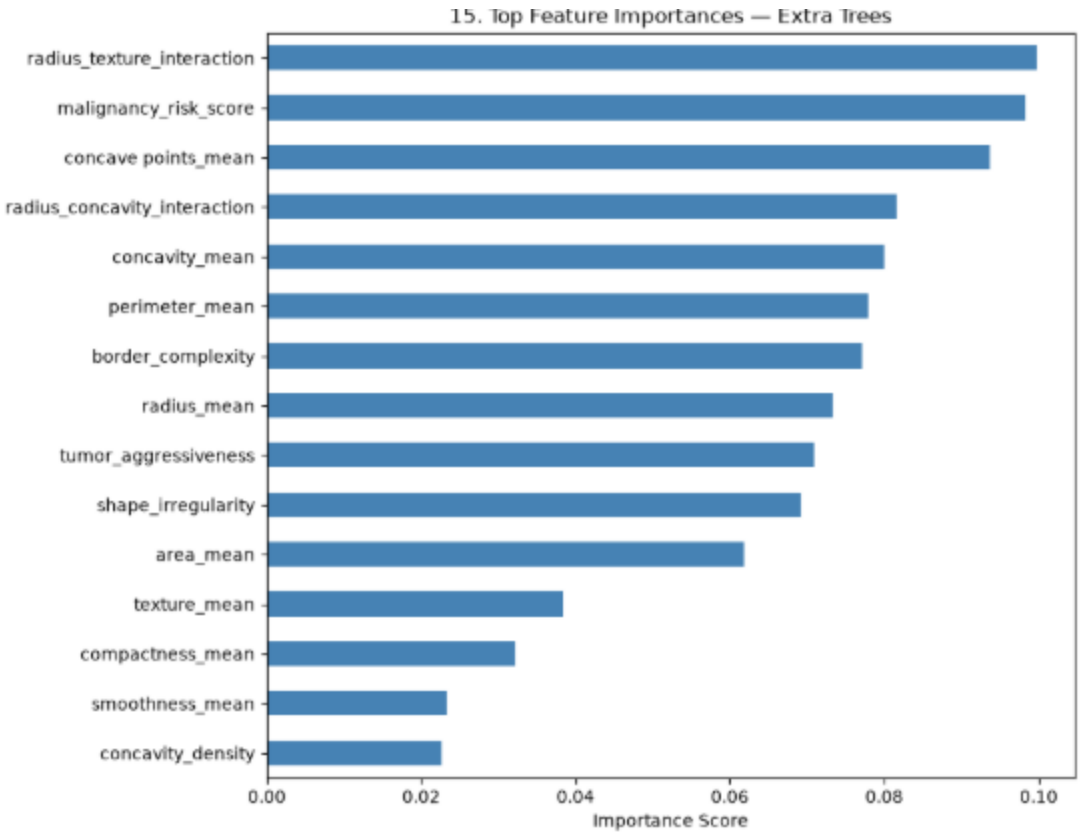
Extra trees recorded the highest test accuracy (99.82%), ROC-AUC of 1.0000 And mean five-fold cross validation accuracy of 100.00%. The Extra Tree classification report rendered precision, recall and F1 score for both classes to 1.00. Rounded values do not prove every prediction was correct, as the confusion metrics should be used to examine the actual errors. Accuracy measures. Overall correct predictions. Precision measures how often predicted positives are correct, while recall measures how many actual positives are identified. F1 score balances precision and recall. ROC_AUC measures ranking ability across thresholds. In a cancer related task, malignant class recall is especially important because false negatives mean malignant cases are classified as benign.

MODEL EVALUATION FIGURES




13. Confusion Matrix — Extra Trees
694
0
2
404
B
M
Actual Diagnosis
Predicted Diagnosis
600
500
400
300
200
100
0


14. ROC Curves — All Models
True Positive Rate
False Positive Rate
Logistic Regression (AUC = 0.9842)
Random Forest (AUC = 0.9999)
Extra Trees (AUC = 1.0000)
Gradient Boosting (AUC = 0.9998)
XGBoost (AUC = 0.9999)



MODEL IMPROVEMENT AND LIMITATIONS

I did 5 model comparisons, but then there's not much clarity on how to show a separate before and after experiment where parameters were changed after diagnosing and weakness. The next step would be to tune extra trees using cross validation on training data, varying parameters such as N_estimators, max_depth, min_samples_leaf and max_features. The test set should remain untouched until final evaluation. The high schools need careful checking. The model should be re evaluated without race scores or variables that may be derived from diagnosis, then tested on an independent data set. These checks are necessary before claiming real world performance.

CONCLUSION AND RECOMMENDATIONS

The EDA showed meaningful differences in tumor size and shape measurements across the diagnosis classes. Extra Trees was the strongest model in the recorded comparison, achieving 99.82% test accuracy and 1.000 ROC-AUC. However, the data sets, provenance and engineered variables must be verified to rule out leakage. The model is promising for this data set but is not ready for clinical use without independent validation.

The recommendation is to verify the data set source and how engineer predictions were created. To repeat model evaluation without any feature that may import diagnosis. To prioritize malignant class, recall and examine false negatives, not just the accuracy. To perform parameter training using training, only cross validation and evaluate ones on an untouched test set. Then lastly is to validate on representative, independent clinical data before considering real world use.